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Floristic diversity along the various altitude of Himachal Pradesh

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Abstract : A journey begins from Ranchi, Jharkhand (651m i.e 2140 feet to Lahaul and Spiti valley of Himachal Pradesh (3080m i.e 10,100 feet)..The journey from Ranchi to Lahaul and Spiti valley was made on two wheels (make -bullet).The prime objective of the tour was to observe and study the changing vegetation of the various altitude. The place which I crossed wereManali(2050m)-Rohtang(3960m)-Spitivalley,Tandi(2573m)-Keylong(3349m)-Jispa(3142m). The ranging vegetation (2050>3960m asl) was amazing and picturous. The photograph were taken and identified in the field as well as in the laborotary from some of the selected sites. In the present paper some of the plants is being highlighted which are changing with altitude and the plants has been discussed and presented.

Keywords : Changing vegetation, Altitude, Above sa level-Asl

INTRODUCTION

Himachal Pradesh, located in the lap of the Himalayas, has varied climatic conditions due to variations in altitude ranging from 450 meters to 6500 meters above mean sea level from west to east and from south to north (Verma, 2014)[1]. It is located between 28° to 33° N AND 75° to 79° E. The state encompasses extraordinary vegetation in communities and floral assemblages, which could be attributed to wide altitudinal gradient coupled with local variations, such as heavy rainfall in the southern parts of the state and arid conditions in the northern part of the state (Singh *et al* 2000)[2].These wide variations in altitude, topography and climate have made this state a home for wide variety of flora and fauna.

MATERIALS AND METHOD

The work was undertaken through field study carried out during the period 10th July 2015 to 25th July 2015 in single season. Photographs were taken of the changing vegetation of Ladakh along the National highway no.1 (N.H-1) and from the nearby valley side. Altitudes were recorded as according to the height of the pass, which were mentioned in the rock cut hoardings. The assessment of species diversity was done by listing the plants species (shrubs and herbs) present within the study area and the identification of the species was confirmed by the experts. The vegetation data was quantitatively analysed for species, habit, status and altitude.

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RESULTS

There were eight plant species distributed at different altitude. Two species were found from the same family Asteraceae but both of them were at different altitude. *Aster himalaicus* were found at ranging altitude of 3600-4500 m with frequent vegetation and *Taraxacum officinale* was very common and abundant in the Himalayan region, often found in meadows and forest of Himachal Pradesh starting from altitude above 1000 m. *Trifolium repens* of family Fabaceae were found abundantly from an altitude of 700 meter. *Viola canescens* of family Violaceae which is also endangered species in some of the places is found at an altitude of 1500 meter -2400 meter with sparse condition. *Peganum harmala* of family Zygophyllaceae were found with sparse condition at an altitude of 3400 meter. *Cindelofolia stylosa* of family Boraginaceae and *Urtica hyperborean* with family Urticaceae both were found at an altitude of 4900 meter with consequently with sparse and frequent condition. *Rhododendron arboreum* of family Ericaceae were found abundantly at an altitude of 2000 meter to 3000 meter showing in Table I.

Table I: Showing plant species with habit, family, status and altitude variation.

Sl.No	Botanical name	Habit	Family	Status	Altitude
1	*Aster himalaicus*	Herb	Asteraceae	Frequent	3600-4500m
2	*Trifolium repens*	herb	Fabaceae	abundantly	>700m
3	*Viola canescens*	Herb	Violaceae	Sparse	1500-2400m
4	*Taraxacum officinale*	Herb	Asteraceae	Widespread	>1000m
5	*Peganum harmala*	Herb	Zygophyllaceae	Sparse	3400m
6	*Cindelofolia stylosa*	Herb	Boraginaceae	Sparse	4900m
7	*Urtica hyperborean*	Herb	Urticulaceae	Frequent	4900m
8	*Rhododendron arboreum*	Shrub or small tree	Ericaceae	Abundant	2000-3000m

DISCUSSION

During the course of study seven species of herbs and one species of trees was recorded. Among these *Viola canscens, Rhododendron arboreum* were highly of highly medicinal value. *Rhododendron arboreum* plants holds the Guinness record for the World largest Rhododendron and is widely popular for its medicinal benefits and economic value and also a postal stamp was issued by the Indian postal department to commemorate this flower, some of the plants like *Taraxum officinale* are referred as phyto food also(Xavier *et al* 2012)[3], some herbs like *Urtica Hyperborea* are rich source of antioxidants(Raj *et al* 2012)[4], and species like *Cindelofolia stylosa* have antimicrobial properties and species *Peganum harmala* have also medical importance (Aslam *et al* , 2014)[5]. *Trifolium repens* have also the medicinal value (Maher. E.P. and M.A. Hyughes, 1973)[6] and beside the medicinal importance *Aster himalaicus* and *Trifolium repens* can be very good ornamental plants if cultivated. So, all the above plant species are valuable resources of Himachal Pradesh and some of the need to be conserved before being rare and endangered by involving research institutes and state departments through establishment of medicinal parks, herbal gardens, ex-situ conservation and in-situ conservation and cultivation will be beneficial for the sustainable utilization of these wild resources in future.

CONCLUSION

Himachal Pradesh vegetation is significant and important asset to our country. Native vegetation along Manali, Rohtang, Lahaul and Spiti valley serves as a natural sustainable and minimal maintenance ground cover which helps prevent soil erosion, reduces water run-off and provides diversity in the landscape .vegetation along these area side and nearby valley provides essential habitat for native flora and fauna of several species .So, it is very important to document the all type of plant species along the valley side and nearby by implementing both with or without the act of Indian Government.

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Fig 1. *Aster himalaicus*



Fig 2. *Trifolium repens*



Fig 3. *Viola canescens*



Fig 4. *Taraxacum officinale*



Fig 5. *Peganum harmala*



Fig 6. *Cindelofolia stylosa*



Fig 7. ***Urtica hyperborea***



Fig 8. ***Rhododendron arboreum***

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